BCA Vth Semester
Computer Communication and Networks
Course Code: BCAC0011

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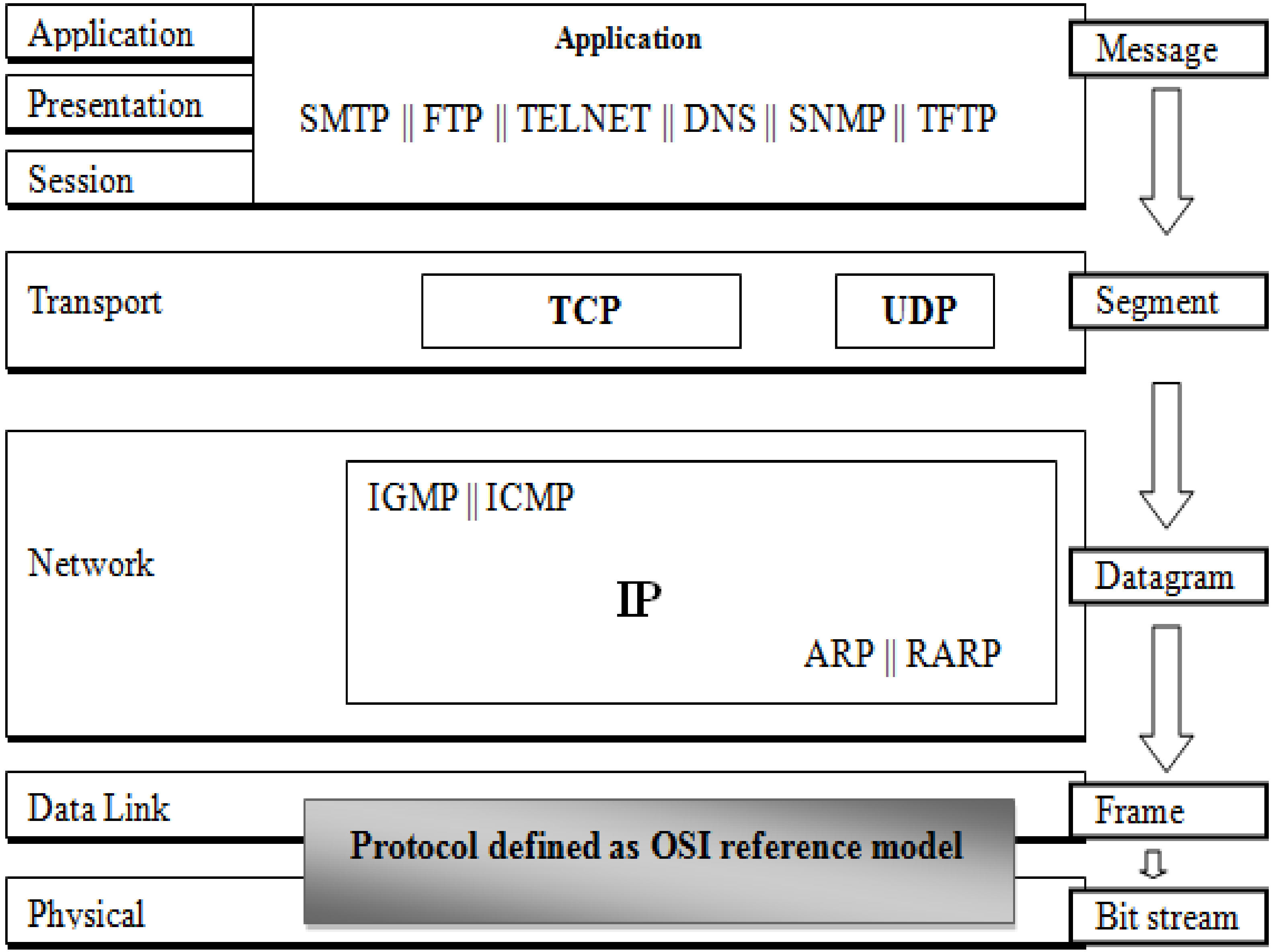
TCP/IP Reference model:

Transmission control protocol/Internetworking protocol (TCP/IP) is a set of protocols that define how all transmissions are exchanged across the internet.
An internet under TCP/IP operates like a single network connecting many computers of any size and type.

TCP/IP and OSI

TCP was developed before the OSI model. Therefore the layers in TCP/IP do not match exactly with those in the OSI model.
The TCP/IP protocol is made of the five layers that are Physical, Data link, Network, Transport and Application. The application layer in TCP/IP can be combination of Session, Presentation and Application layers of the OSI model.

Figure: *TCP/IP and OSI reference model*



Network Layer:

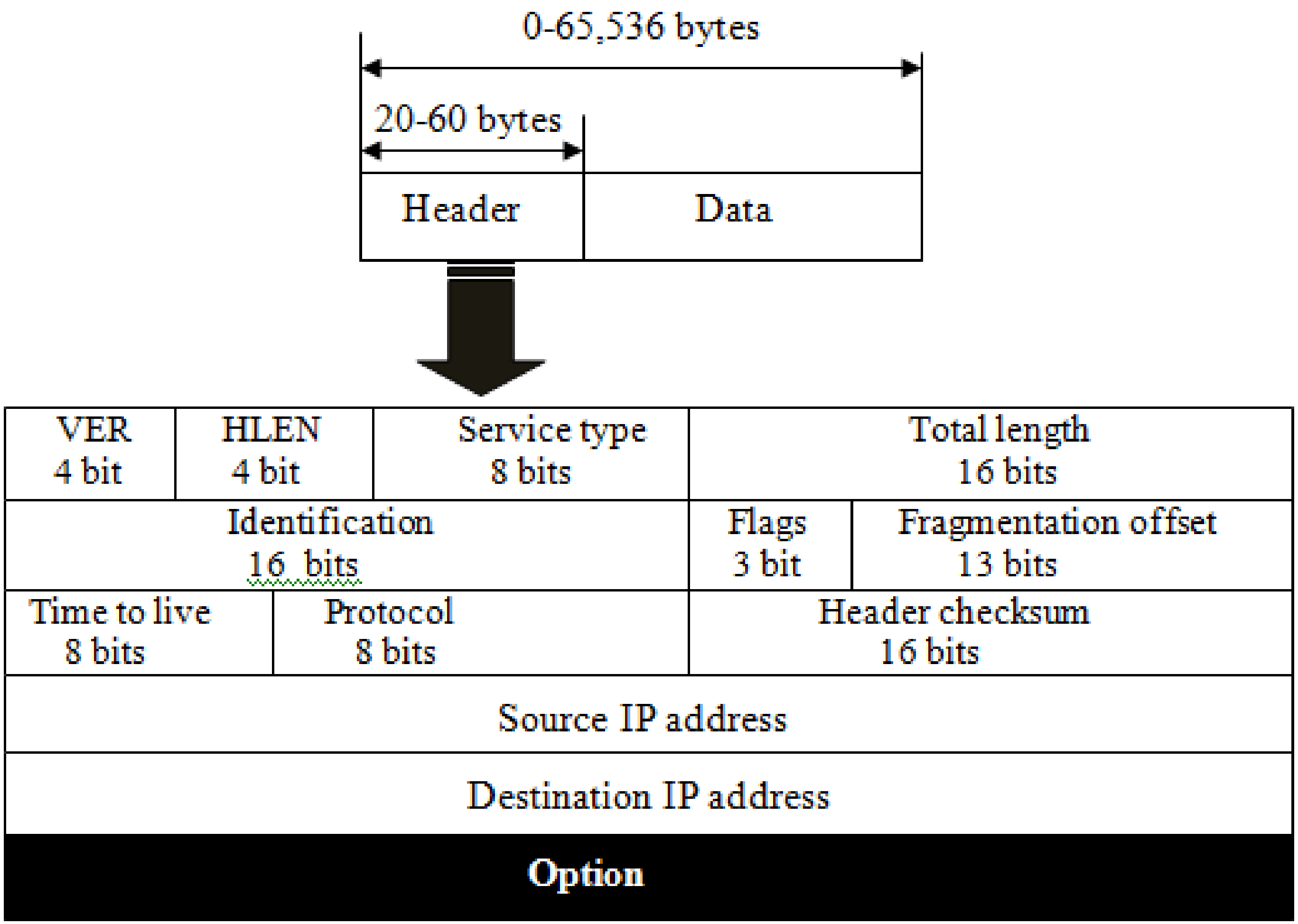
Internet Protocol (IP) IP is the transmission mechanism used by the TCP/IP protocols. It is an unreliable and connection less protocol with no error checking or tracking. It transmits the data to the destination but with no guarantees. If reliability is important, IP must be paired with a reliable protocol such as TCP.

IP transmit the data in the form of packet called datagram. Each datagram transmit separately and may travel along different routes. It may arrive out of order or duplicate. IP does not keep track of the routes.

Datagram:

Packets in the IP layer are called datagram. A datagram is a variable length packet (up to 65,536 bytes) consisting of two parts: header and data. The header can be from 20 to 60 bytes and contains information essential to routing and delivery. A brief description of header is

Figure: *ID datagram*



1. **Version:** This field defines the version number of the IP. The current version is 4 **(IP_V 4).**

2. Header length (HLEN): The HLEN field defines the length of the header.

3. Type Of Service (ToS): This helps the router in taking the right routing decisions. The structure is:

First three Bit : They specify the precedence i.e. the priority of the packets.

Next three bits:

D bit - D stands for delay. If the D bit is set to 1, then this means that the application is delay sensitive, so we should try to route the packet with minimum delay.

T bit - T stands for throughput. This tells us that this particular operation is throughput sensitive.

R bit - R stands for reliability. This tells us that we should route this packet through a more reliable network.

Last two bits: The last two bits are never used.

4. Total Length: The total length field defines the total length of the IP datagram.

5. Identification: The identification field is used in fragmentation. A datagram passing through different networks may be dividing in to fragments to match the network frame size. When this happen each fragment is identified with a sequence number in this field.

6. Flag: It has three bits -

M bit : If M is one, then there are more fragments on the way and if M is 0, then it is the last fragment

DF bit: If this bit is sent to 1, then we should not fragment such a packet.

Reserved bit: This bit is not used.

7. Fragmentation offset: It is a 13 bit field that represents where in the packet, the current fragment starts.

8. Time To Live (TTL) : Using this field, we can set the time within which the packet should be delivered or else destroyed. It is strictly treated as the number of hops. The packet should reach the destination in this number of hops. Every router decreases the value as the packet goes through it and if this value becomes zero at a particular router, it can be destroyed.

9. Protocol: This specifies the module to which we should hand over the packet (UDP or TCP).

10. Header Checksum: This is the usual checksum field used to detect errors.

11. Source: It is the IP address of the source node.

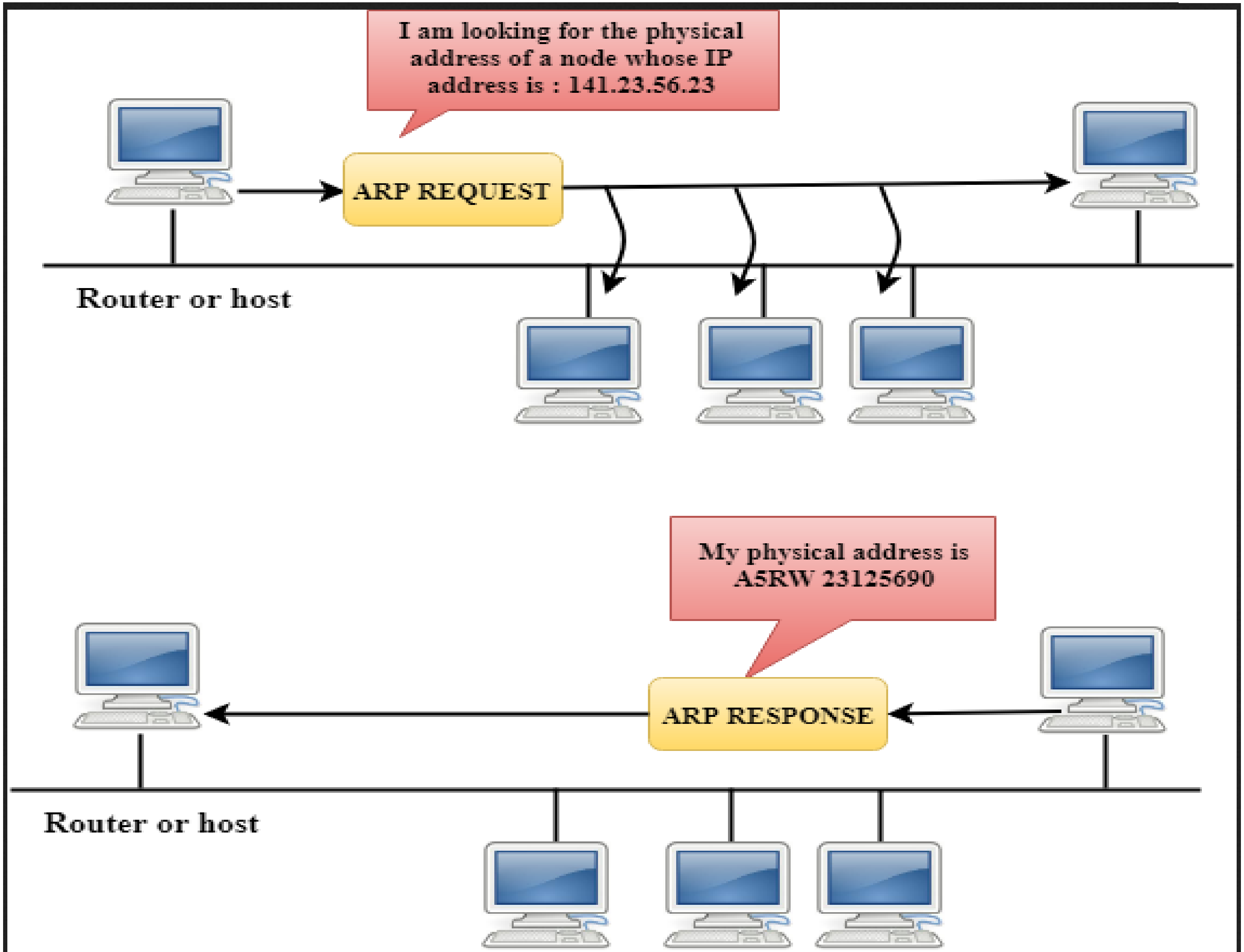
12. Destination: It is the IP address of the destination node.

13. IP Options: The options field was created in order to allow features to be added into IP as time passes and requirements change.

ARP

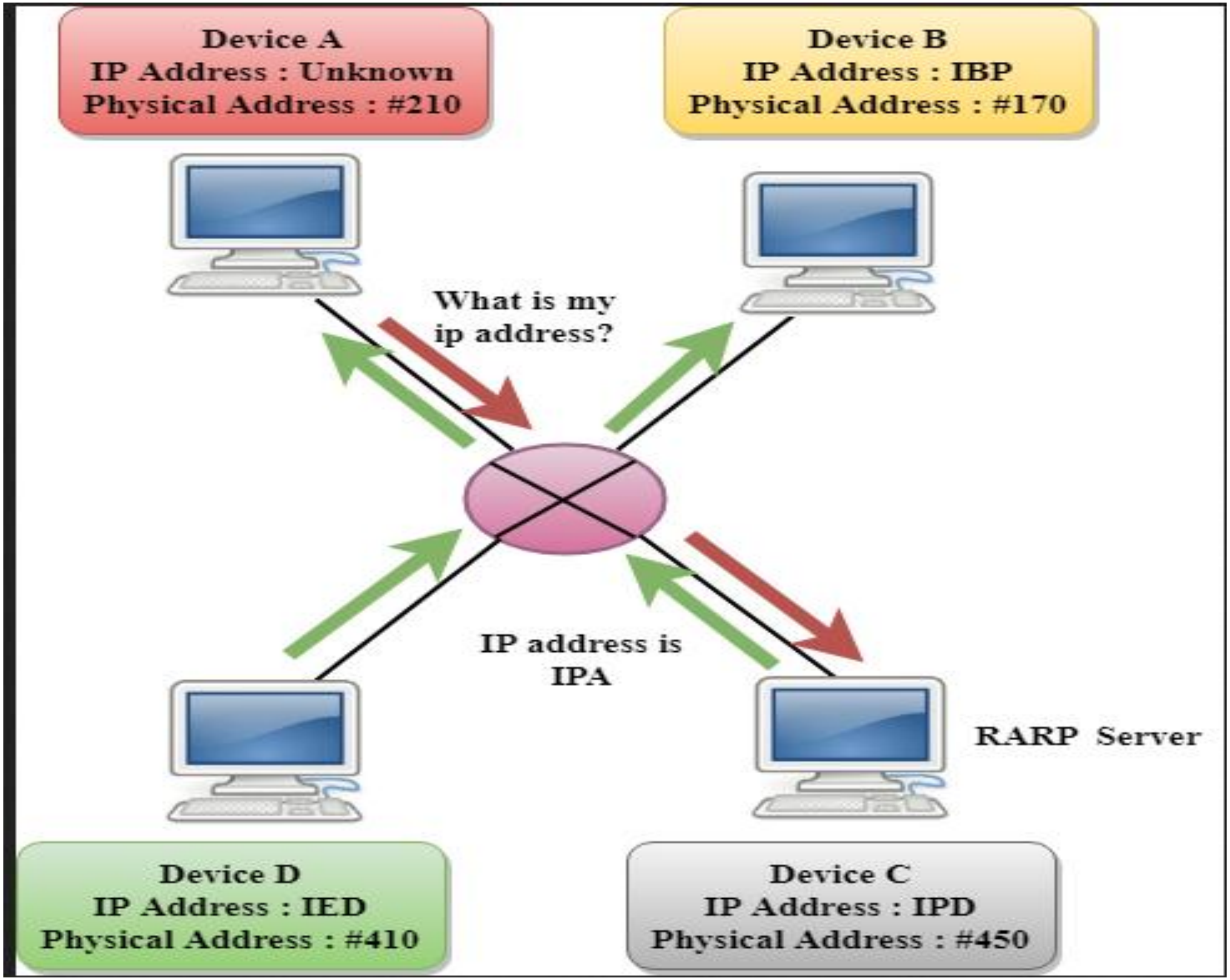
- ARP stands for Address Resolution Protocol.
- It is used to associate an IP address with the MAC address.
- Each device on the network is recognized by the MAC address imprinted on the NIC. Therefore, we can say that devices need the MAC address for communication on a local area network. MAC address can be changed easily but IP address does not change. ARP is used to find the MAC address of the node when an internet address is known.

If the host wants to know the physical address of another host on its network, then it sends an ARP query packet that includes the IP address and broadcast it over the network. Every host on the network receives the ARP packet, but only the intended recipient recognizes the IP address and sends back the physical address. The host holding the datagram adds the physical address to the cache memory and to the datagram header, then sends back to the sender.


I am looking for the physical address of a node whose IP address is : 141.23.56.23
ARP REQUEST
Router or host
My physical address is A5RW 23125690
ARP RESPONSE
Router or host

RARP

RARP stands for **Reverse Address Resolution Protocol**.
If the host wants to know its IP address, then it broadcast the RARP query packet that contains its physical address to the entire network. A RARP server on the network recognizes the RARP packet and responds back with the host IP address.
The protocol which is used to obtain the IP address from a server is known as **Reverse Address Resolution Protocol**.
The message format of the RARP protocol is similar to the ARP protocol.
Like ARP frame, RARP frame is sent from one machine to another encapsulated in the data portion of a frame.


Device A
IP Address : Unknown
Physical Address : #210
Device B
IP Address : IBP
Physical Address : #170
What is my
ip address?
IP address is
IPA
RARP Server
Device D
IP Address : IED
Physical Address : #410
Device C
IP Address : IPD
Physical Address : #450

ICMP (Internet Control Message Protocol)

- The ICMP is a network layer protocol used by hosts and routers to send the notifications of IP datagram problems back to the sender.
- ICMP uses echo test/reply to check whether the destination is reachable and responding.
- ICMP handles both control and error messages, but its main function is to report the error but not to correct them.
- An IP datagram contains the addresses of both source and destination, but it does not know the address of the previous router through which it has been passed. Due to this reason, ICMP can only send the messages to the source, but not to the immediate routers.
- ICMP messages are transmitted within IP datagram.

Five types of errors are handled by the ICMP protocol:

i) Destination unreachable: The message of "Destination Unreachable" is sent from receiver to the sender when destination cannot be reached, or packet is discarded when the destination is not reachable.

ii) Source Quench: The purpose of the source quench message is congestion control. The message sent from the congested router to the source host to reduce the transmission rate.

iii) Time Exceeded: Time Exceeded is also known as "Time-To-Live". It is a parameter that defines how long a packet should live before it would be discarded.

iv) Parameter problems: When a router or host discovers any missing value in the IP datagram, the router discards the datagram, and the "parameter problem" message is sent back to the source host.

v) Redirection: When the host sends the datagram to a wrong router. The router that receives a datagram will forward a datagram to a correct router and also sends the "Redirection message" to the host to update its routing table.

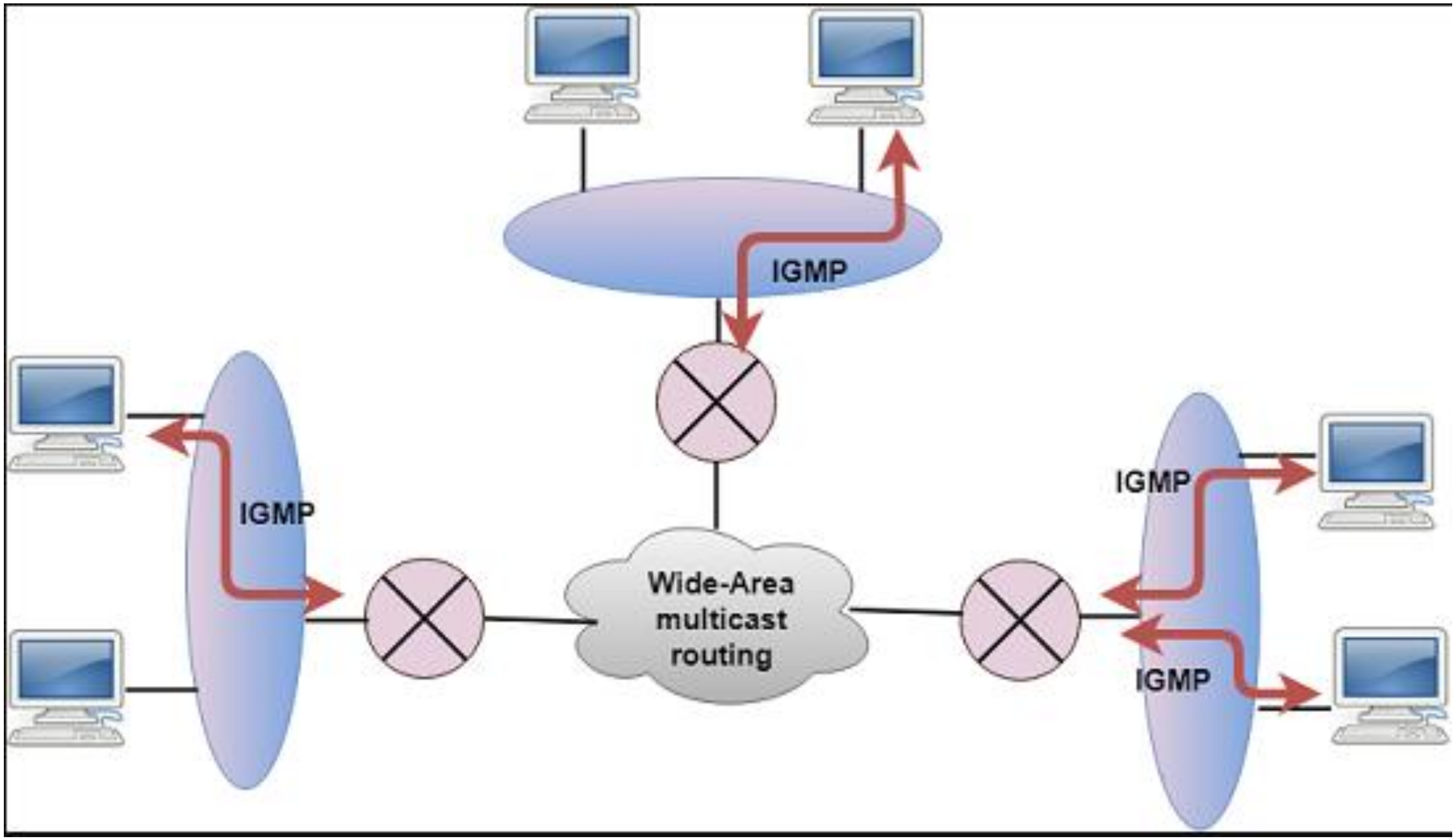
IGMP (Internet Group Message Protocol)

The IP protocol supports two types of communication:

- **Unicasting:** It is a communication between one sender and one receiver. Therefore, we can say that it is one-to-one communication.
- **Multicasting:** Sometimes the sender wants to send the same message to a large number of receivers simultaneously. This process is known as multicasting which has one-to-many communication.

The IGMP protocol is used by the hosts and router to support multicasting.

The IGMP protocol is used by the hosts and router to identify the hosts in a LAN that are the members of a group.


IGMP
IGMP
IGMP
IGMP
Wide-Area multicast routing

Transport Layer:

Traditionally transport layer was represented in TCP/IP by two protocols: TCP and UDP. IP is a **host to host** protocol, means that it can deliver a packet from one physical device to another.

UDP and TCP are transport level protocol responsible for delivery of a message form process (running program) to another process.

Addressing:

At the transport layer, we need a transport layer address, called a port number, to choose among multiple processes running on the destination host. The destination port number is needed for delivery; the source port number is needed for reply.

In the internet model, the port numbers are 16- bit integers between 0 to 65,535. The client program defines itself with a port number, chosen randomly by the transport layer software running on the client host.

It should be clear by now that the IP addresses and port numbers play different roles in selecting the final destination of data. The destination IP address defines the host among the different hosts in the world. After the host has been selected, the port number defines one of the processes on this particular host.

IANA Ranges:

The **IANA** (Internet Assigned number Authority) has divided the port numbers in to three ranges: well known, registered and dynamic (or private).

Well Known Ports: The ports ranging from 0 to 1023 are assigned and controlled by IANA use by the application end points that communicate using the Internet's Transmission Control Protocol (TCP) or the User Datagram Protocol (UDP). These are well known ports. For example, a remote job entry application has the port number of 5; the Hypertext Transfer Protocol (HTTP) application has the port number of 80, for DNS the port number is 53.

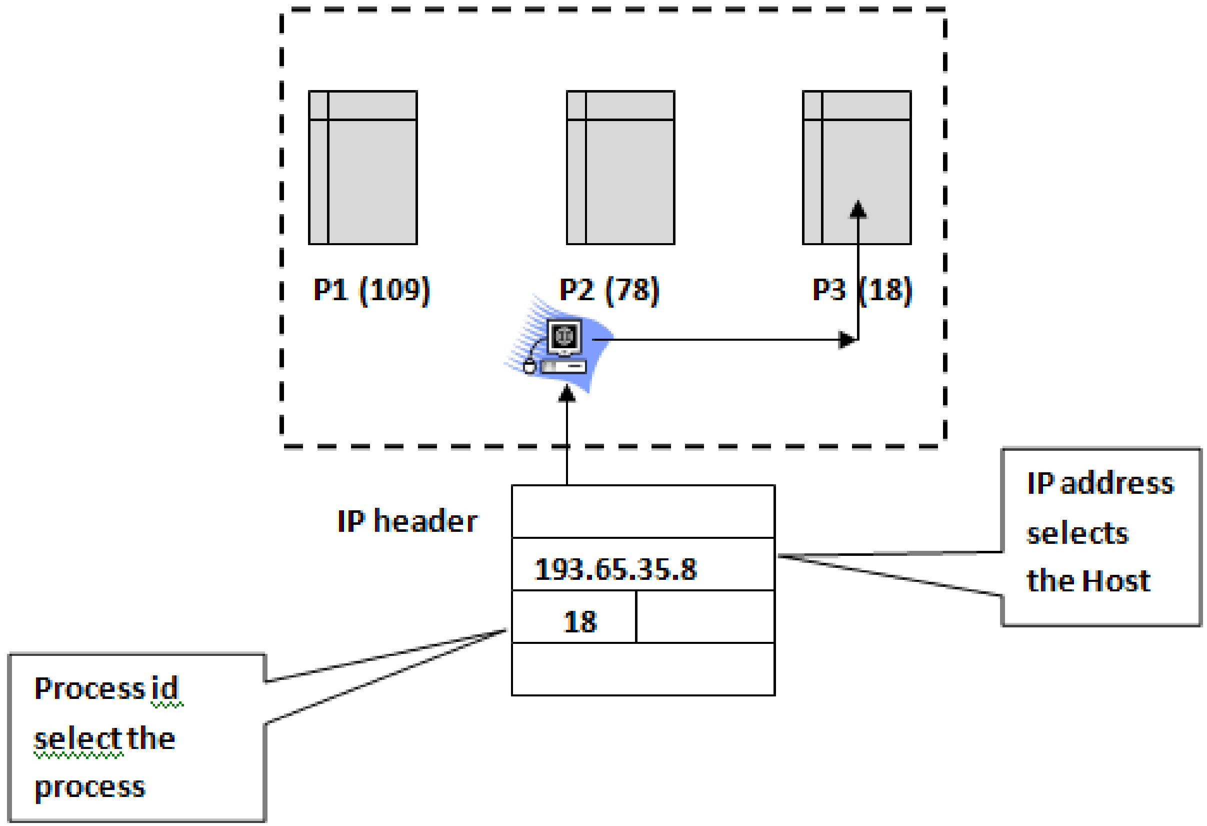
Registered Ports: The port ranging from 1024 to 49151 are assigned by IANA for specific service upon application by a requesting entity. 4224 is used for Cisco Audio Session Tunneling

Dynamic Ports (private port number): the port ranging from 49,152 to 65,535 are available for use by any application to use in communicating with any other application, using the Internet's Transmission Control Protocol (TCP) or the User Datagram Protocol (UDP). These are momentary ports.

Socket Address:

Process to process delivery needs two different identifiers, IP address and the port number at each end to make a connection. The combination of an IP address and port number is called a socket address. The client socket address defines the client process uniquely and server socket address defines the server process uniquely.

IP address versus Port Address

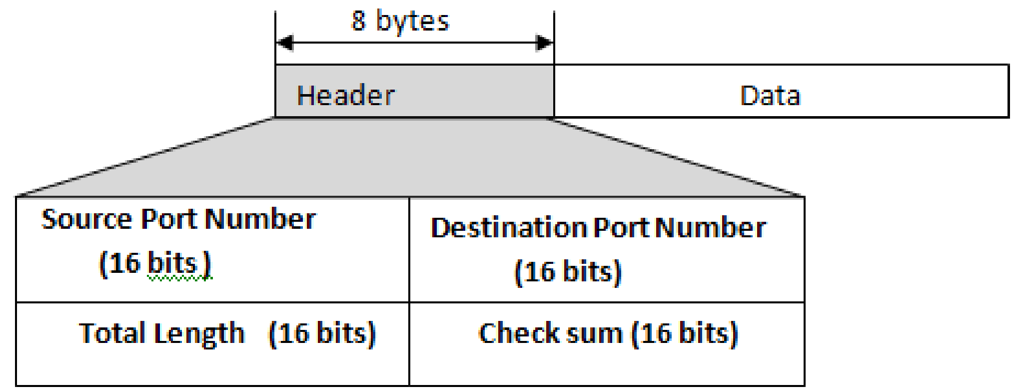


User Datagram protocol:

- The user datagram protocol (UDP) is called conn unreliable transport protocol.
- It does not add anything to the service of IP except to provide process-to-process communication instead of host–to-host communication.
- It performs very limited error checking.
- UDP is very simple protocol using a minimum of overhead.
- If a process wants to send a small message and does not care much about reliability, it can use UDP.
- Sending a small message by using UDP takes much less interaction between the sender and receiver then using TCP.

User Datagram:

UDP packets, called user datagrams, have a fixed –size header of 8 bytes.



Source port Address:

This is the port number used by the process running on the source host. It is 16 bit long, which means that the port number can range from 0 to 65,535.

Destination Port address:

This is the port number used by the process running on the destination host. It is 16 bit long, which means that the port number can range from 0 to 65,535?

Total Length:

This is a 16 bit field that define a total length of the user datagram **header** and **data.** The 16 bits can define the total length of 0 to 65,535 bytes.

The length field in the user data gram is not necessary. A user datagram is encapsulated in an IP datagram. There is a field in the IP datagram that defines the total length. There is another field in the IP datagram that defines the length of the header. So if we subtract the value of the second field from the first, we can deduce the length of a UDP datagram that is encapsulated in an IP datagram.

UDP length= IP length- IP header's Length

Transmission Control Protocol (TCP):

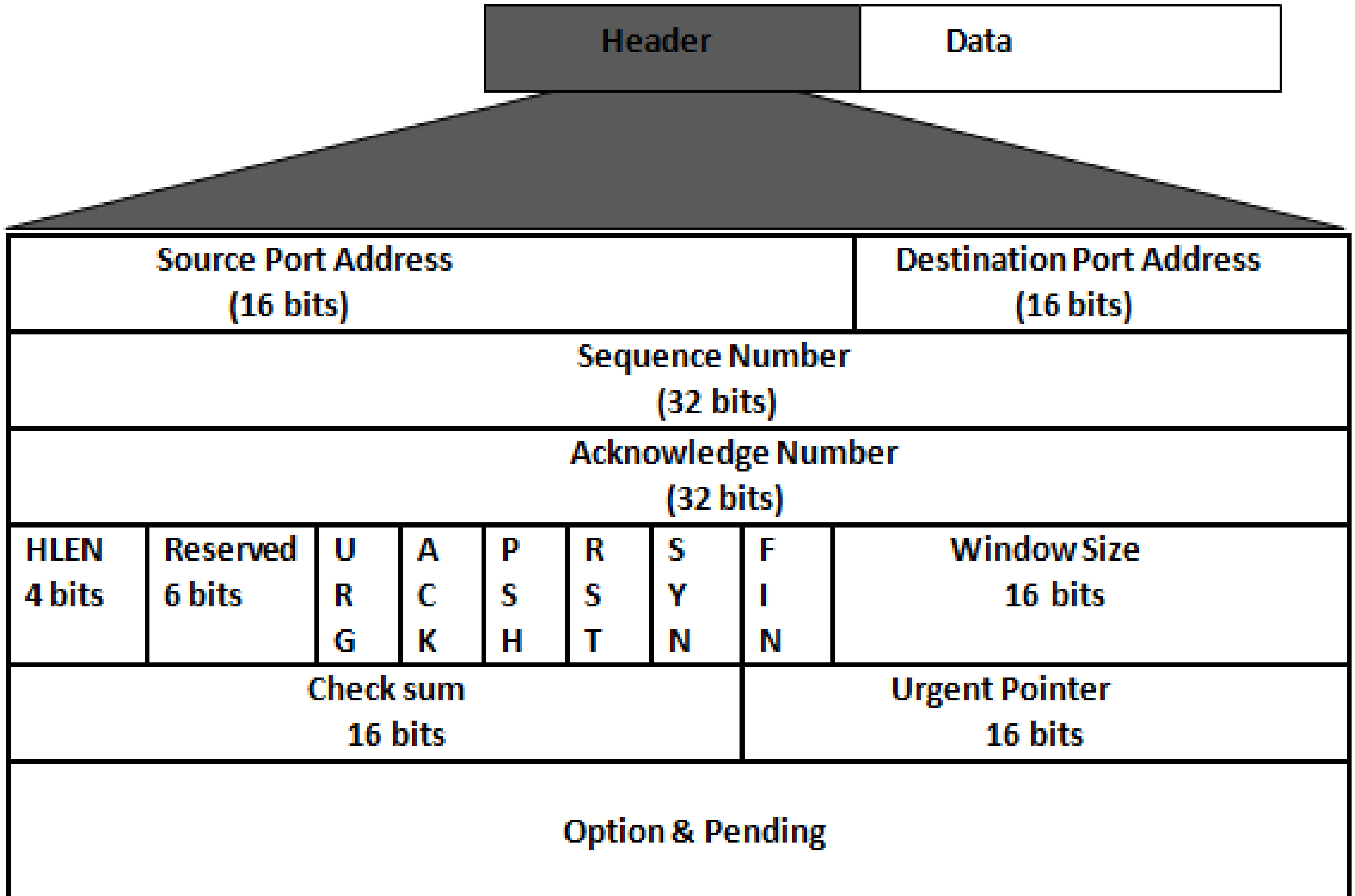
- TCP is also a Process-to-process communicating protocol.
- Unlike UDP, TCP is a connection oriented protocol.
- Like UDP, it also uses the port number.
- It creates a virtual connection between two TCPs to send data.

TCP Services:

- **Process-to-process communication**
- **Stream Delivery Services**
- **Sending and receiving buffers**
- **Segments**
- **Full duplex Communication**
- **Connection oriented Services**
- **Acknowledgement**
- **Error Control**
- **Flow Control**
- **Congestion Control**

Segment:

TCP carries the data in the form of segment.TCP provides the reliable transmission but less in speed in comparison to UDP. Because of its smaller frame size the UDP is much faster than TCP. A brief description of each fields are here



Source Port Address: The source port address define the application program in the source computer .

Destination Port Address: The destination port address defines the application program in the destination computer.

Sequence Number: A stream of data from the application program may be divided in two or more TCP segments. The sequence number defines the position of the data in the original data stream.

Acknowledgement Number: The 32 bit acknowledgement number is used to acknowledge the receipt of data from the other communicating device. This number is valid only if the ACK bit in the control field is set.

Header Length (HLEN): The four bit HLEN field indicates the number of 32- bit (four byte) words in the TCP header.

Reserved: This is a 6 bit field reserved for future use.

Window Size: This field define the size of the window, in bytes, that the sender and receiver must maintain the size of window.

Checksum: This 16 bit field contains the checksum. Check sum is mandatory in TCP.

Control: This field defines 6 different control bits or flag as shown in figure these are:

- **URG:** The value of the urgent pointer field is valid.
- **ACK:** The value of the acknowledgement field is valid
- **PSH:** Push the data
- **RST:** Reset the connection
- **SYN:** Synchronize sequence Number during connection
- **FIN:** Terminate the connection

Urgent Pointer: The 16 bit field, which is valid only if the urgent flag is set. This field is used when the segment contains urgent data.

Options: There can be up to 40 bytes of optional information in the TCP Header.

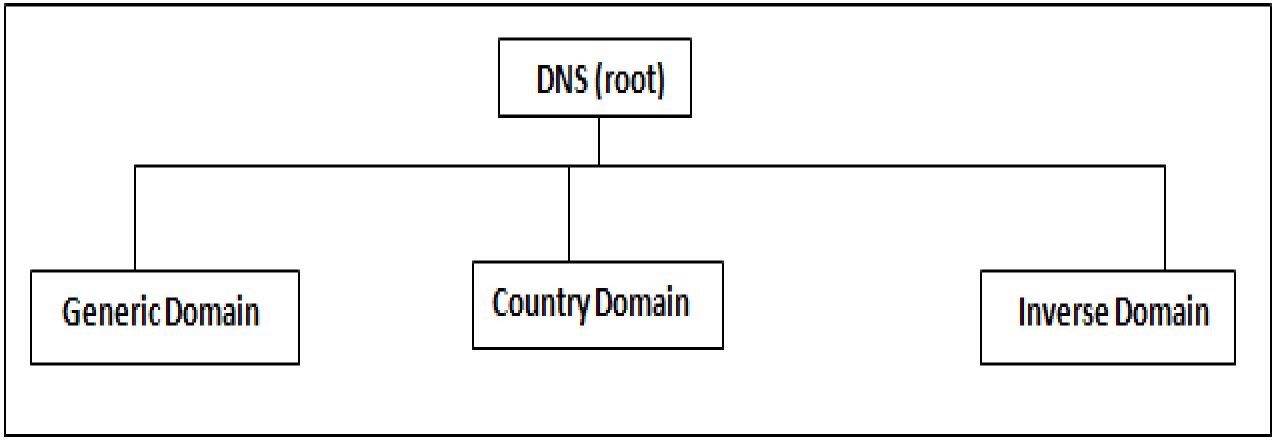
Domain Name System (DNS):

To identify an entity, TCP/IP protocols use the IP address, which uniquely identifies the connection of a host to the internet. However people prefer to use names instead of addresses. Therefore we need a system that can map a name to an address and conversely an address to a name. In TCP/IP this is the **Domain Name System (DNS).**

DNS in the Internet:

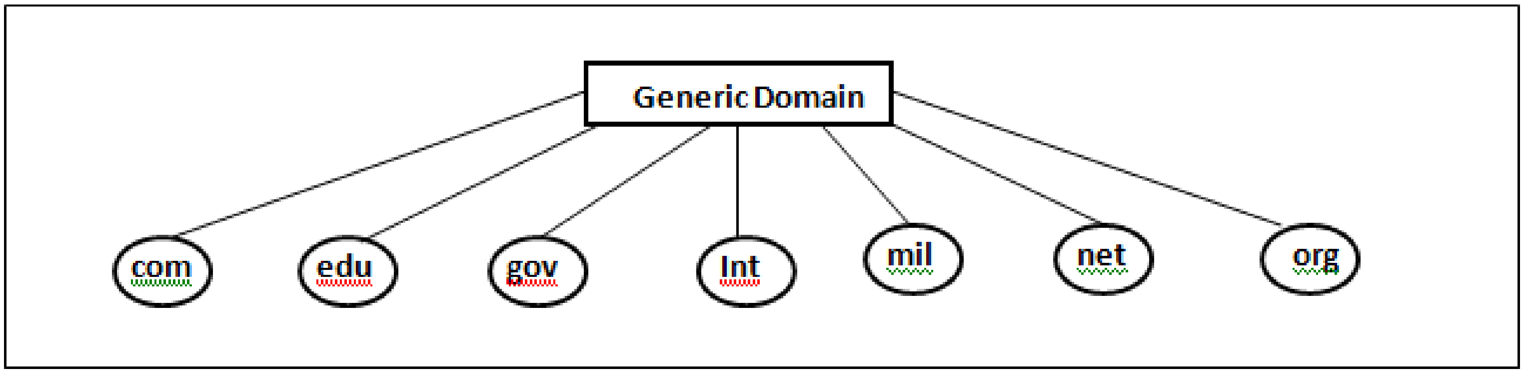
DNS is a protocol that can be used in different platforms. In the Internet the domain name space (tree) is divided into three different sections:

- **Generic Domain**
- **Country Domain**
- **Inverse Domain**



The generic domains define registered hosts according to their generic behaviour. The first level in the generic domain section allows seven possible three-character labels. These labels describe the organization typ

Generic Domain



Generic Domain Labels:

Label	Description
com	Commercial Organization
edu	Educational Institutions
gov	Government institutions
int	International organization
mil	Military Group
net	Network support Centers
org	Nonprofits Organizations

Country Domains:

The country domain section follows the same format as the generic domains country but uses two-character country abbreviations (i.e. “in” for India, “us” for United States). Second level can be organizational, or they can be more specific, national designations.

i.e. The address “du.delhi.in” can be translated to Delhi University in Delhi in India.

Inverse Domain:

Inverse domain is used to map an address to a name. This may happen, for example when a server has received a request from a client to do a task. Whereas the server has a file that contains a list of authorized client, the server lists only the IP address of the client (extract from the received IP packet). To determine if the client is on the authorised list, it can send a query to the DNS server and ask for a mapping of address to name.

TELNET:

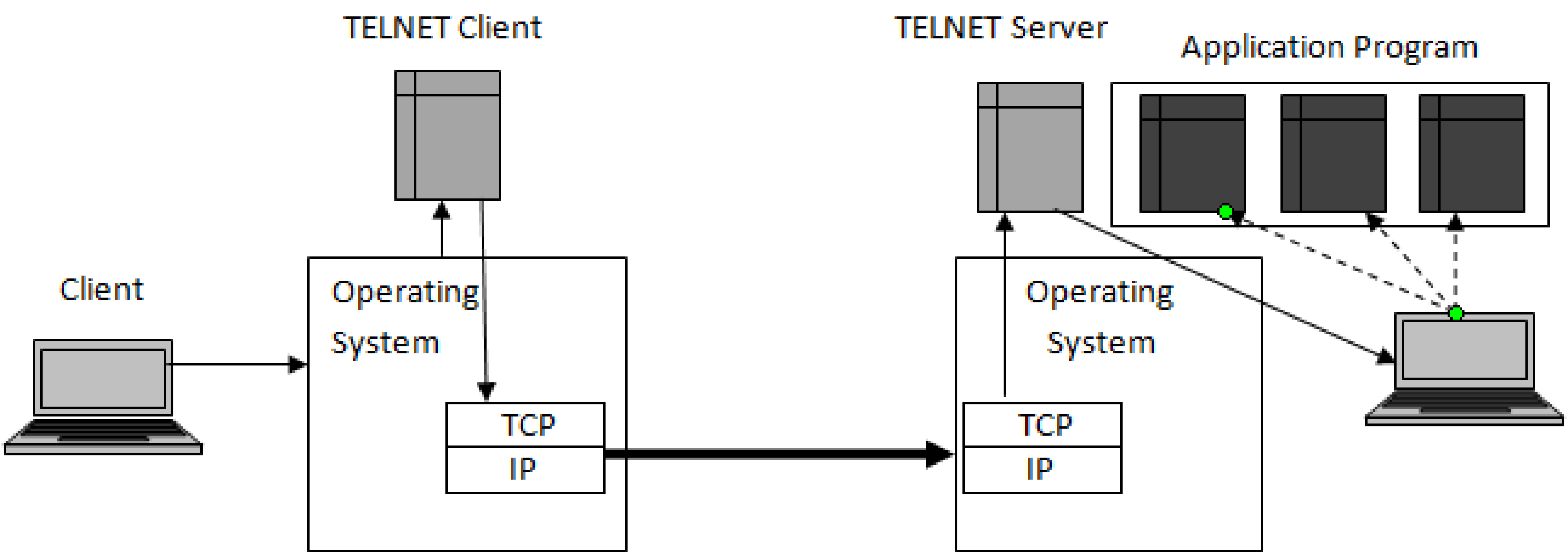
The main task of the Internet and its TCP/IP protocol suite is to provide services for users. For example user wants to be able to run different application programs at a remote site and create results that can be transferred to their local site.

The better solution is a general purpose client-server program through which user can access any application program on a remote computer. After logging on, a user can use the services available on the remote computer and transfer the results back to the local computer.

For this purpose there is a popular client-server application program called TELNET. TELNET means TErminaL NETwork. TELNET enables the establishment of a connection to a remote system in such a way that the local terminal appears to be a terminal as the remote system.

Remote Login:

When a user wants to access an application program or utility located on a remote machine, can be possible through **remote login.** Here the TELNET client server program comes in to use. The user send the keystrokes to the terminal driver where the local operating system accepts the but does not interpret them the character are sent to the TELNET client, which transforms the characters to a universal character set called **network virtual terminal character** and delivers them to server.



File Transfer Protocol

- FTP (File Transfer Protocol) is the simplest and most secure way to exchange files over the Internet.
- Transferring files from a client computer to a server computer is called **"uploading"** and transferring from a server to a client is **"downloading".**
- To access an FTP server, users must be able to connect to the Internet or an intranet (via a modem or local area network) with an FTP client program.

File Transfer Protocol

- FTP doesn’t not really move, it copies files from one computer to another
- FTP is the file transfer protocol in the Internet's TCP/IP protocol suite’s Application Layer.

File Transfer Protocol

- An FTP Client is software that is designed to move files back-and-forth between two computers over the Internet.
- It needs to be installed on your computer and can only be used with a live connection to the Internet.

Features

- **The FTP protocol is used for transferring one file at a time**, in either direction, between the client machine (the one which initiated the connection, i.e. the calling machine) and the server machine (which provided the FTP service, i.e. the called machine).
- The FTP protocol can also perform other actions, such as creating and deleting directories (only if they are empty), listing files, deleting and renaming files, etc.

Features

- FTP allows files to have ownership and access restrictions
- FTP hides the details of individual computer systems

Example

- downloading MP3.
- Online games rely heavily on FTP.
- Other events like auctions and online trading use FTP to transact business
- **FTP** is commonly used to transfer Web page files from their creator to the computer that acts as their server for everyone on the Internet.

Differences between FTP and HTTP

- The major difference between FTP and HTTP is that FTP is a two-way system - FTP can be used t copy or move files from a server to a client computer as well as upload or transfer files from a client to a server.
- HTTP, on the other hand, is strictly one-way: "transferring" text, pictures and other data(Multimedia files) from the "server" to a client computer which uses a web browser to view the data.

Differences between FTP and HTTP

- FTP systems generally encode and transmit their data in binary sets which allow for faster data transfer; HTTP systems encode their data in MIME format which is larger and more complex.
- Files are automatically copied or moved from a file server to a client computer's hard drive, and vice versa. On the other hand, files in an HTTP transfer are viewed and can 'disappear' when the browser is turned off unless the user executes commands to move the data to the computer's memory.

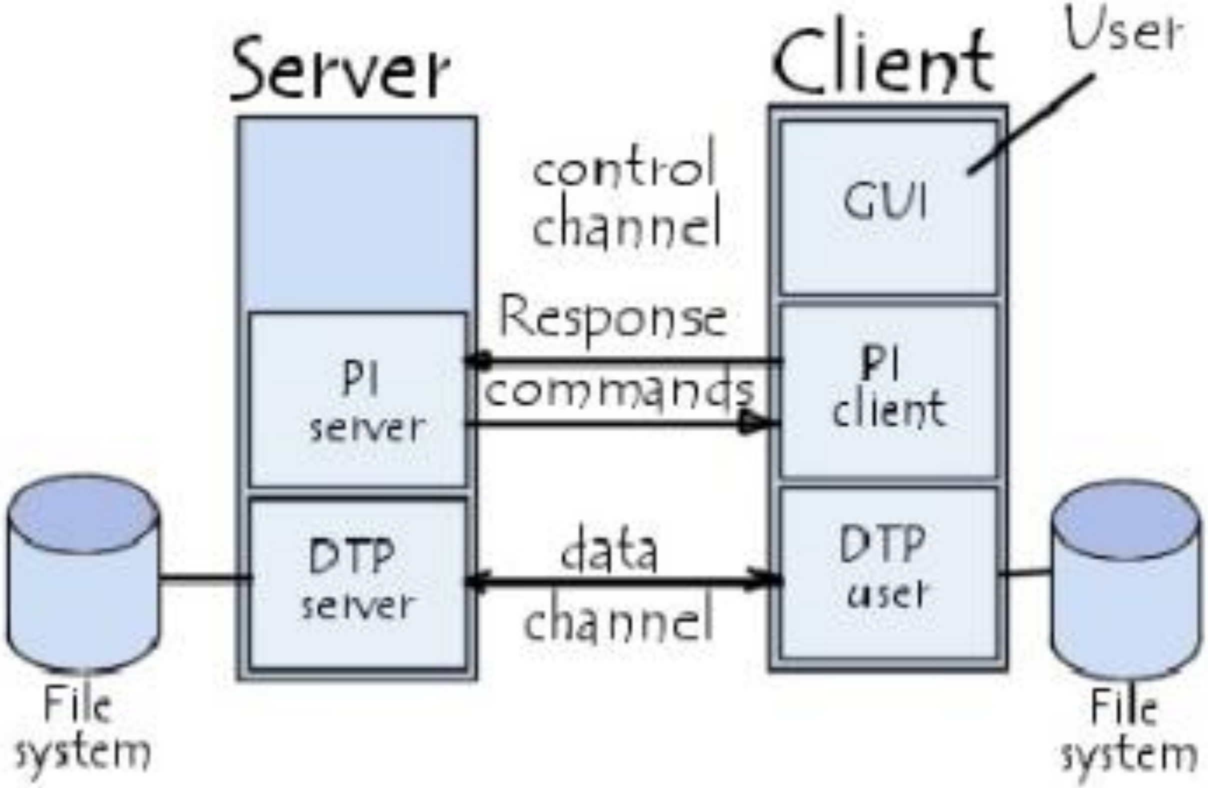
The FTP model

FTP protocol falls within a client-server model, i.e. one machine sends orders (the client) and the other awaits requests to carry out actions (the server).

During an FTP connection, two transmission channels are open:

- A channel for commands (control channel) , a control channel that stays open for the entire session
- A channel for data , data channel that opens and closes to transfer data such as folder listings and files to or from the server as requested by the client.

The FTP model



The FTP model

So, both the client and server have two processes allowing these two types of information to be managed:

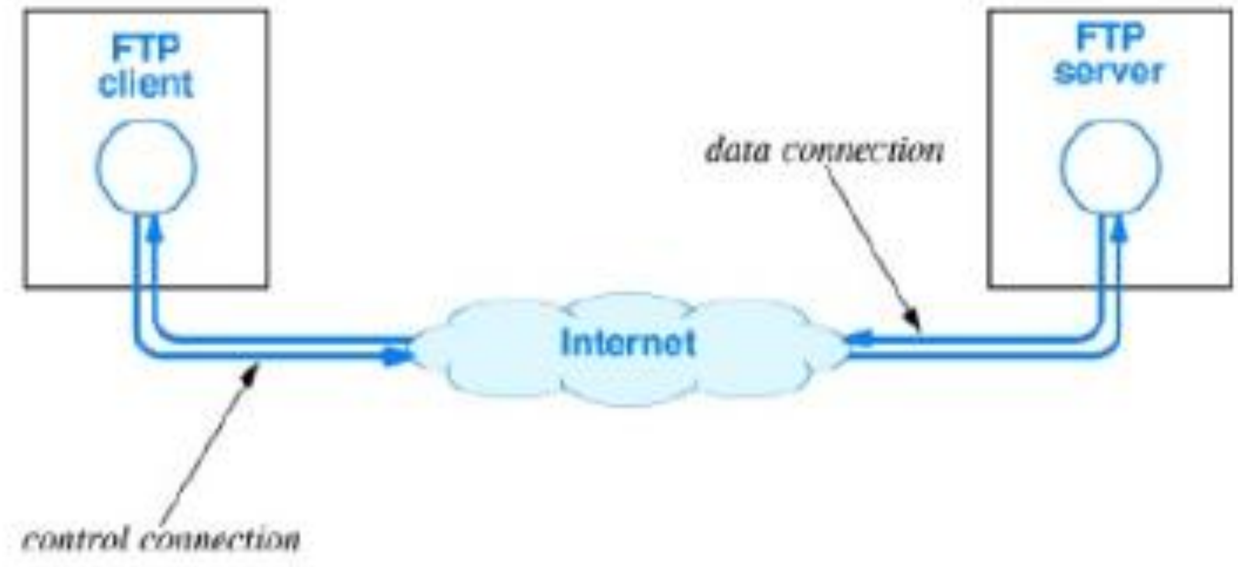
- **DTP** (*Data Transfer Process*) It establishes the connection and managing the data channel. The server side DTP is called *SERVER-DTP*, the client side DTP is called *USER-DTP*
- **PI** (*Protocol Interpreter*) Controls DTP using commands received over the control channel. It is different on the client and the server:

The FTP model

- The **SERVER-PI** is responsible for listening to the commands coming from a USER-PI over the control, establishing the connection for the control channel, receiving FTP commands from the USER-PI over this, responding to them and running the SERVER-DTP.
- The **USER-PI** is responsible for establishing the connection with the FTP server, sending FTP commands, receiving responses from the SERVER-PI and controlling the USER-DTP if needed.

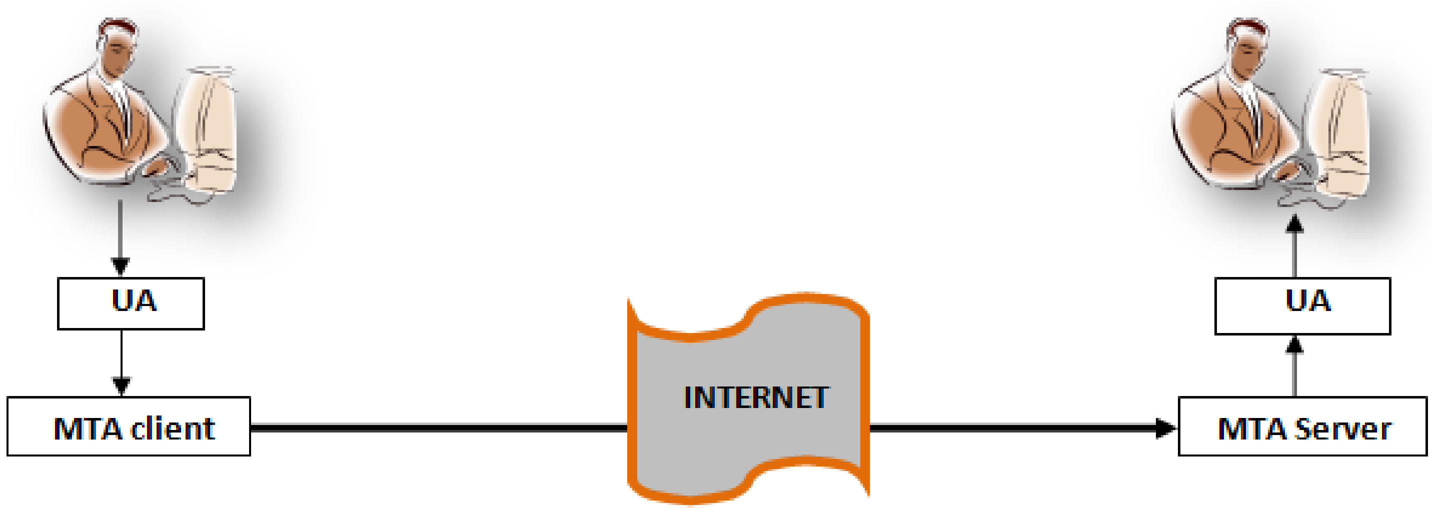
The FTP model

- When an FTP client is connected to a FTP server, the USER-PI initiates the connection to the server according to the Telnet protocol.
- The client sends FTP commands to the server, the server interprets them, runs its DTP, then sends a standard response.
- Once the connection is established, the server-PI gives the port on which data will be sent to the Client DTP.
- The client DTP then listens on the specified port for data coming from the server.


FTP
client
FTP
server
data connection
Internet
control connection

Simple Mail Transfer Protocol (SMTP)

One of the most popular network services is **electronic mail (e-mail).** The TCP/IP protocol that supports electronic mail on the Internet is called **Simple mail transfer Protocol (SMTP).**



It is a system for sending messages to other computer users based on e-mail address. SMTP provides for mail exchange between users on the same or different computers and supports:

- Sending a single message to one or more recipients.
- Sending message that include text, voice, video and graphics.

SMTP breaking down in two components:

User Agent (UA)

Mail Transfer Agent (MTA)

User Agent (UA):

UA prepares the message, creates the envelope and puts the message in to the envelop. The UA is normally a program used to send and receive mail. To send the message UA use the address of recipients.

Address:

To deliver mail, a mail handling system must use a unique addressing system. The addressing system used by SMTP consists of two parts:

a **local part**

domain name separate by an @ sign.

Local Part:

The local part defines the name of a special file, called the user mailbox, where all of the mail received for a user is stored and retrieval by user agent.

Domain Name:

The second part of the address is the domain name. An organization usually selects one or more hosts to receive and send e-mail.

Mail Transfer Agent (MTA):

The actual mail transfer s done through mail transfer agents (MTAs). To send mail, a system must have a client MTA, and to receive mail, a system must have server MTA.

Multipurpose Internet Mail Extensions (MIME):

SMTP is a simple mail transfer protocol, but SMTP can send message only in seven bit ASCII format. In other words, it has some limitations. For example, it cannot be used for languages that are not supported by seven bit ASCII characters (such as French, German, Russian, Chinese and Japanese). It cannot be used binary files or to send video and audio data.

Post Office Protocol:

SMTP expects that the mail server receiving the mail, to be online all the time, otherwise a TCP connection cannot be established. For this reason, it is not practical to establish an SMTP session with desktop computers because desktop computer are usually powered down at the end of the day.

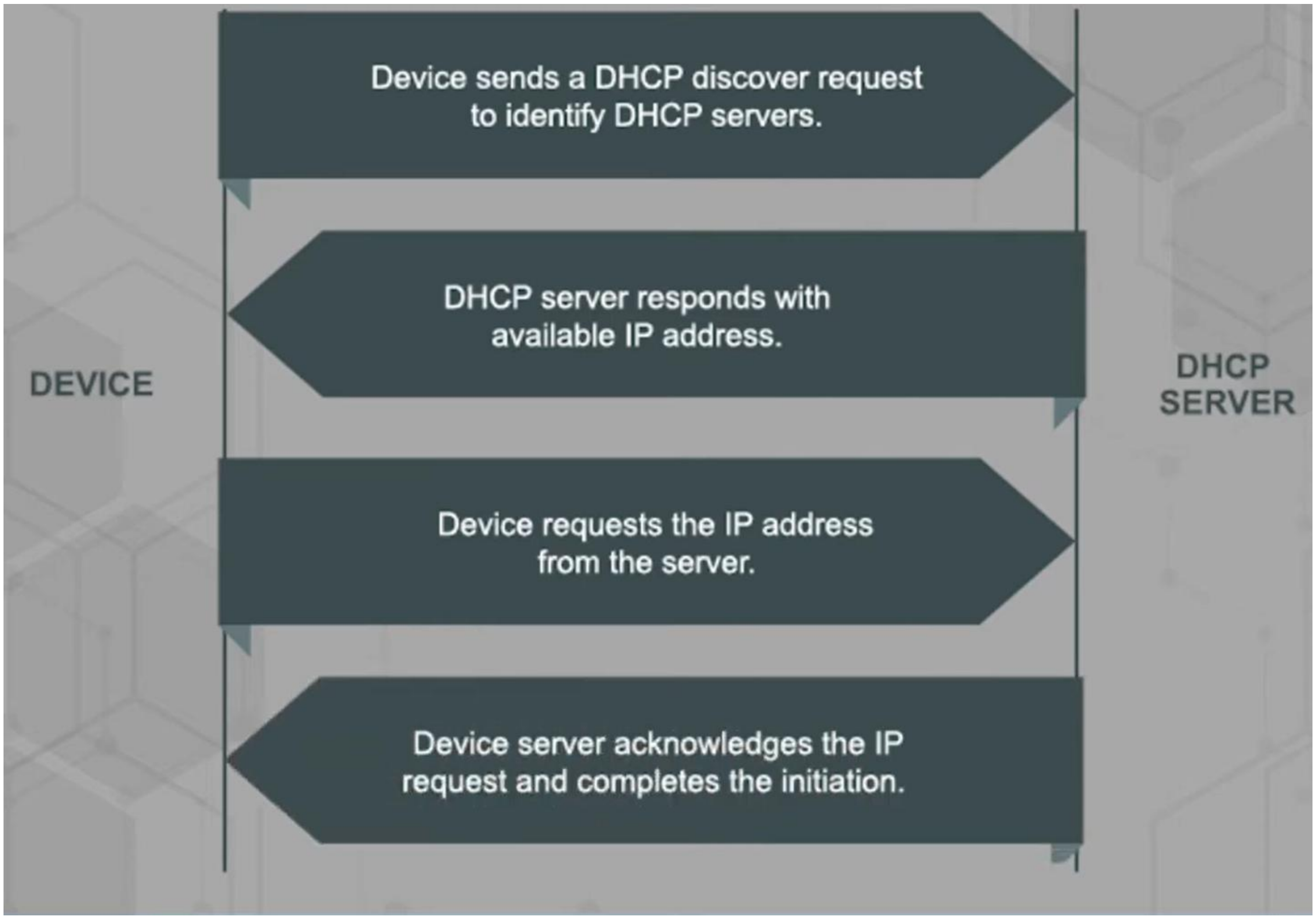
In many organizations, mail is received by an SMTP server that is always on-line. This SMTP server provides a mail drop service. The server receives the mail on behalf of every host in the organization. User workstations interact with the SMTP server to retrieve messages by using client-server protocol such as **Post Office Protocol (POP).**

Although the POP is used to download messages from the server, the SMTP client is needed on the desktop to forward messages from the workstation user to its SMTP mail server.

DHCP (Dynamic Host Configuration Protocol)

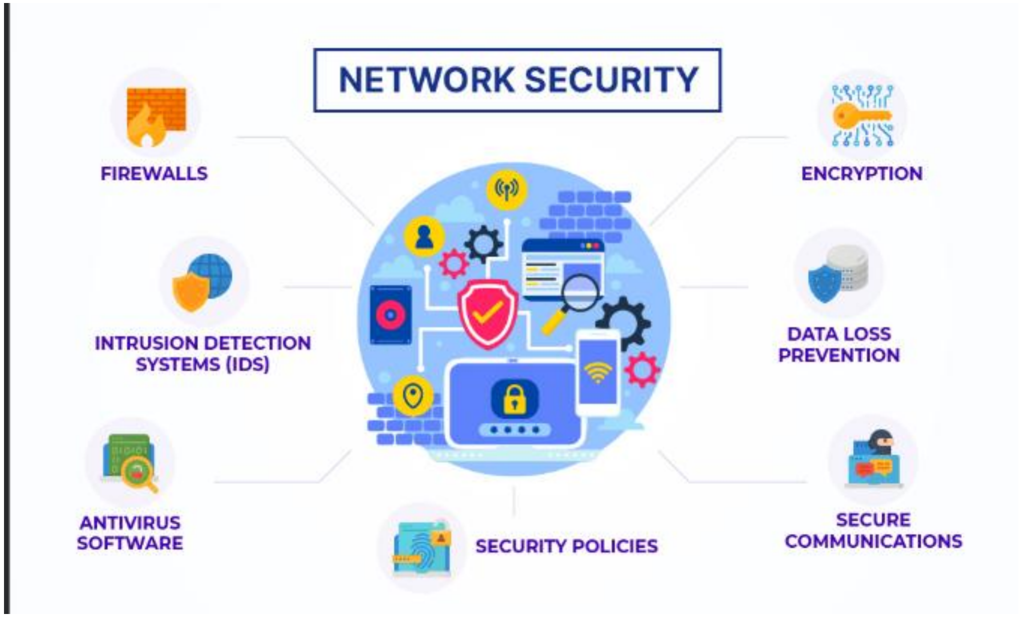
- It is used to dynamically assign an Internet Protocol (IP) address to any device on a network so it can communicate.
- DHCP automates and centrally manages these configurations rather than requiring network administrators to assign IP addresses manually to all network devices.
- Small local networks and large enterprise networks can both implement DHCP.
- DHCP assigns new IP addresses in each location when devices move to a new location on the network.
- DHCP is a client-server protocol.
- Servers manage a pool of unique IP addresses and information about client configuration parameters. The servers assign addresses from those address pools.

DHCP (Dynamic Host Configuration Protocol)

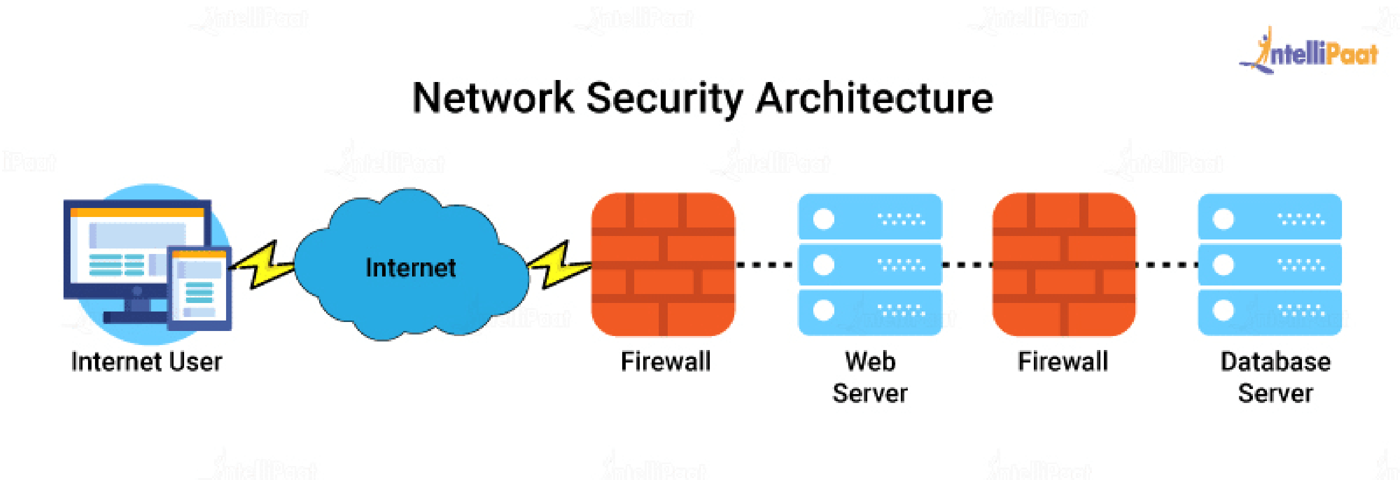


Network security

encompasses the practices and technologies designed to protect the integrity, confidentiality, and availability of computer networks and the data they carry.



Network security



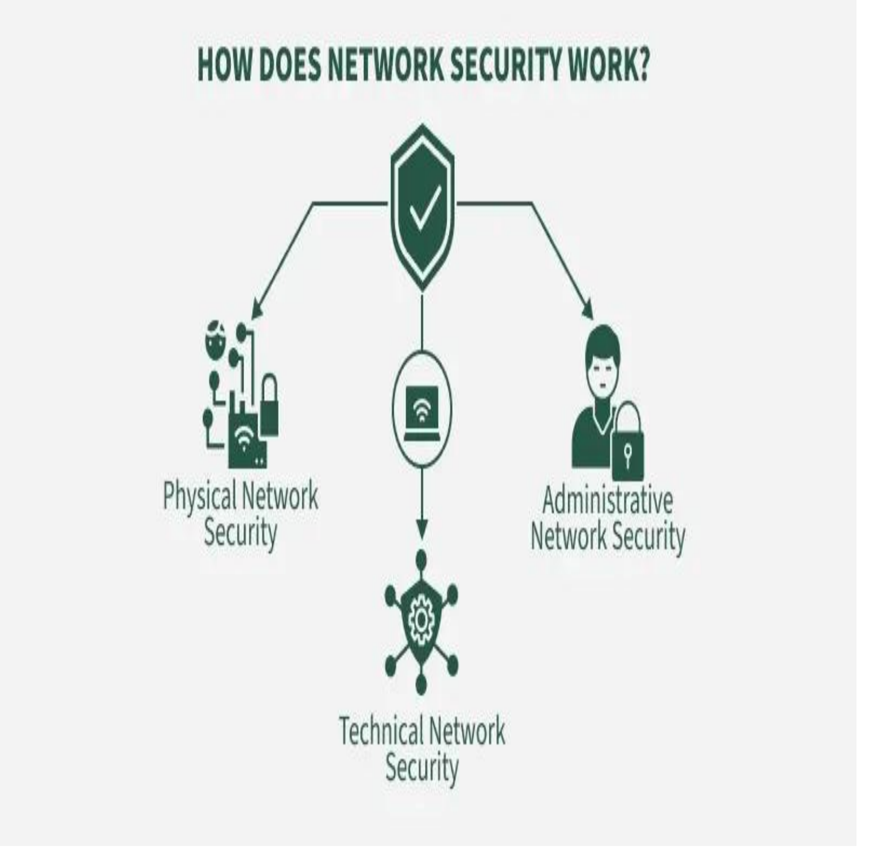
Physical Network Security: It is the foundational level of network protection, focused on preventing unauthorized individuals from gaining physical access to networking equipment and compromising the confidentiality of the network. This can be achieved through measures such as biometric systems, access cards, and other physical security devices.

Technical Network Security: It primarily focuses on protecting the data stored in the network or data involved in transitions through the network. This type serves two purposes. One is protected from unauthorized users, and the other is protected from malicious activities.

Administrative Network Security: This level of network security protects user behavior like how the permission has been granted and how the authorization process takes place. This also ensures the level of sophistication the network might need for protecting it through all the attacks. This level also suggests necessary amendments that have to be done to the infrastructure.

Network security uses multiple layers of protection, both at the network edge and within the internal environment. Each layer enforces specific rules and controls to regulate access to network resources. Authorized users can safely utilize the network, while malicious actors and threats are blocked from causing harm.

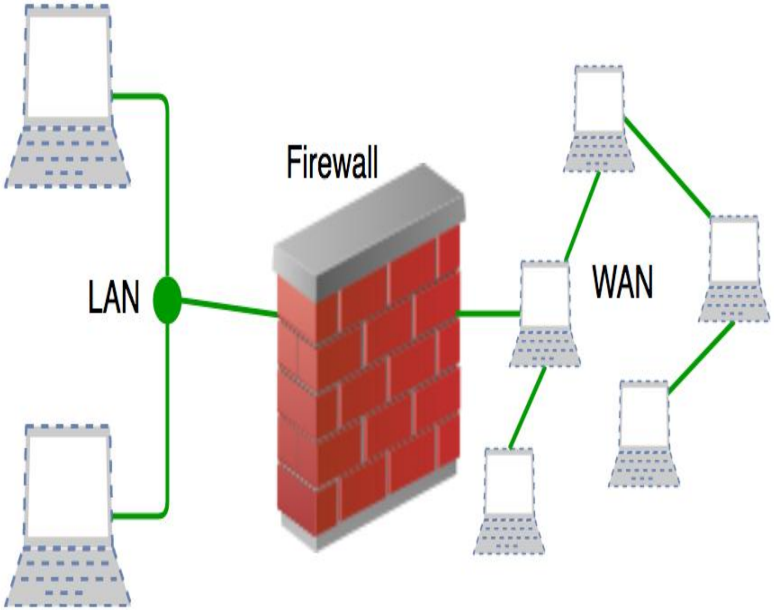
The basic principle of network security is protecting huge stored data and networks in layers that ensure the enforcement of rules and regulations that have to be acknowledged before performing any activity on the data.



FIREWALL

A firewall is a network security system, available as hardware or software, that monitors and controls incoming and outgoing traffic based on predefined rules. It acts like a security guard, filtering data packets to either:

- **Accept:** Allow the traffic.
- **Reject:** Block with an error response.
- **Drop:** Block silently without response.



Firewalls are essential because they:

Prevent Unauthorized Access: Like a locked door with a guard, only trusted users and traffic are allowed through.

Block Malicious Traffic: Harmful data such as viruses, phishing attempts, or denial-of-service (DoS) attacks are stopped before reaching the system.

Protect Sensitive Information: Safeguards personal and business data from theft or accidental leaks.

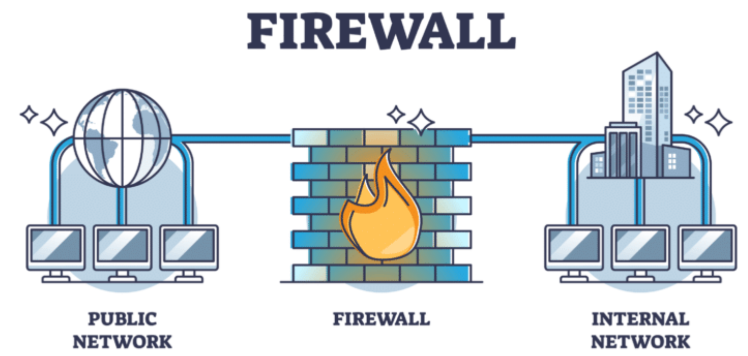
Control Network Usage: Enforces policies such as parental controls, workplace restrictions, or government filtering.

Mitigate Insider Risks: Detects suspicious applications or data exfiltration attempts from within the network.

Working of Firewall

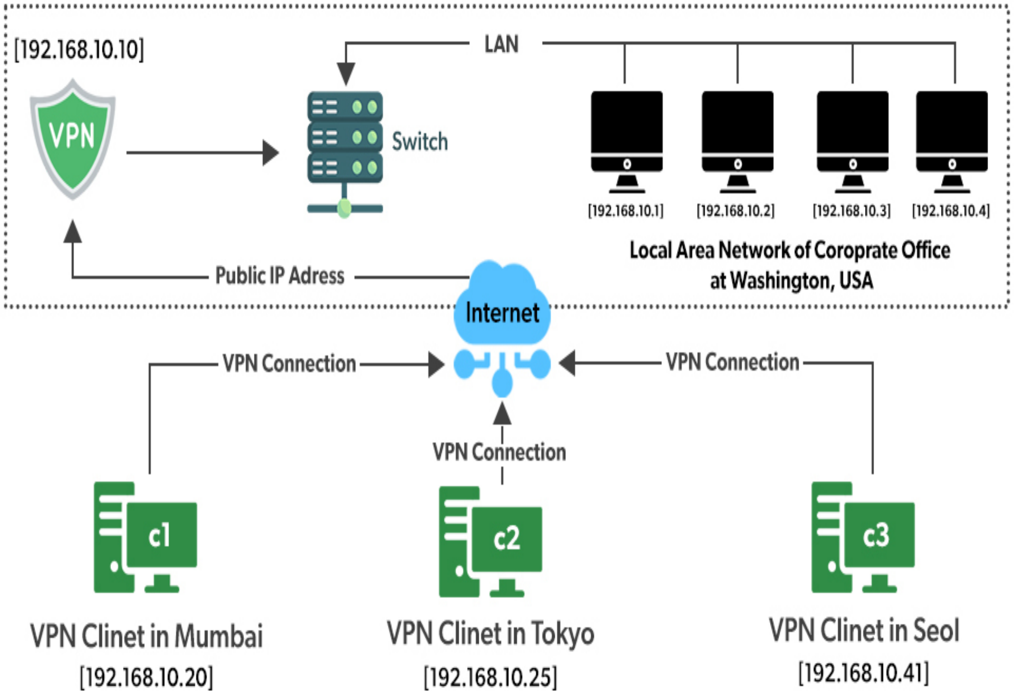
A firewall inspects all incoming and outgoing traffic and decide whether to allow or block it.

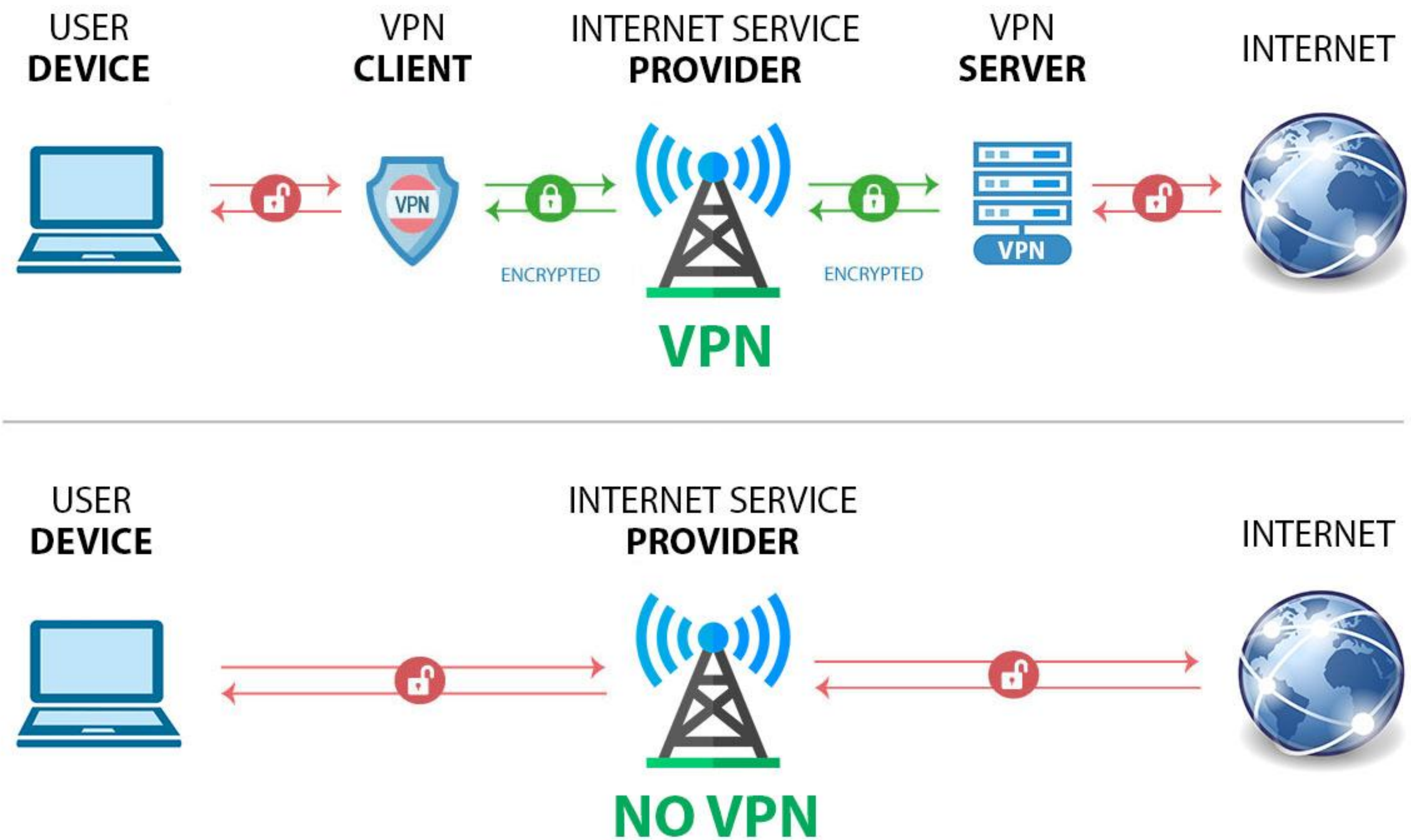
All data packets entering or leaving the network must first pass through the firewall.
The firewall examines each packet against predefined security rules set by the organization.
If the packet matches safe rules, it is allowed; if it is suspicious, blacklisted, or contains malicious content, it is blocked.
Blocked or unusual traffic is recorded in logs, and real-time alerts may be generated for serious threats.
Since it is not possible to define every rule, the firewall applies a default policy (accept, reject, or drop). Setting the default policy to drop or reject is considered best practice to prevent unauthorized access.



A VPN (Virtual Private Network)

A VPN (Virtual Private Network) is a technology that creates a secure, encrypted connection between your device and the internet. It acts as a private tunnel for your internet traffic, preventing unauthorized access and ensuring your online activities remain private. VPNs are essential for enhancing online privacy, protecting sensitive data, and enabling secure access to the internet, especially on public networks.




USER
DEVICE
VPN
CLIENT
INTERNET SERVICE
PROVIDER
VPN
SERVER
INTERNET
VPN
ENCRYPTED
ENCRYPTED
VPN
VPN
USER
DEVICE
INTERNET SERVICE
PROVIDER
INTERNET
NO VPN

How it works

A VPN hides your IP address by letting the network redirect it through a specially configured remote server run by a VPN host. This means that if you surf online with a VPN, the VPN server becomes the source of your data. This means your Internet Service Provider (ISP) and other third parties cannot see which websites you visit or what data you send and receive online. A VPN works like a filter that turns all your data into "gibberish". Even if someone were to get their hands on your data, it would be useless.

Types

Remote Access VPN: This type allows users to connect to a private network from a remote location. It is commonly used by employees to access company resources securely while away from the office.

Site-to-Site VPN: Often used by large organizations, this VPN connects multiple networks, allowing different office locations to communicate securely.

Mobile VPN: Designed for mobile users, this VPN enables secure connections to private networks through cellular networks, ensuring data protection while on the go.

SSL VPN: This type uses the SSL protocol to secure connections, allowing remote users to access private networks through a web browser without needing additional software

VPN functions:

Encryption: A VPN encrypts your internet traffic, converting it into unreadable code while in transit. Only your VPN provider's server and your device can decrypt this data.

IP Address Masking: Your real IP address is replaced with the IP of the VPN server. This hides your identity and location from websites and trackers.

Secure Server Routing: Your data is routed through one of the VPN provider's secure servers before it reaches its destination. This adds an extra layer of security, especially on public Wi-Fi.

Key Security Benefits of Using a VPN

Data Encryption
VPNs use powerful encryption protocols such as OpenVPN, IKEv2/IPsec, or WireGuard to ensure your data is secure. This is especially vital when using unsecured networks like public Wi-Fi in cafes, airports, or hotels, where cybercriminals often lurk.

Anonymous Browsing
When you connect to a VPN, your browsing activity becomes much harder to trace. Your ISP, government agencies, or malicious third parties can't monitor what you're doing online. This protects your privacy and ensures a more anonymous internet experience.

Protection from Hackers
VPNs help prevent cyberattacks like man-in-the-middle (MITM) attacks by encrypting your traffic. Even if a hacker intercepts your data, they won't be able to read it due to encryption.

Bypassing Geo-Restrictions and Censorship
VPNs allow access to content that may be blocked in certain countries. For example, streaming services, social media platforms, or news websites may have region-based restrictions that a VPN can help bypass securely.

How Does TCP Work in Computer Networks?

The TCP working process can be simplified into 3 simple steps:

1. Connection Establishment: TCP establishes a connection between the client and server using a three-way handshake.

2. Data Transfer: Data is broken into smaller chunks called segments and transmitted over the network. During transmission, TCP manages flow control and uses acknowledgment (ACK) flags to confirm successful delivery of data.

3. Connection Termination: After the data transfer is complete, the connection is terminated using a four-way handshake.

TCP Connection Establishment with Three-Way Handshake Process

TCP is a connection-oriented protocol, which means a secure connection is established between two hosts before sending and receiving the actual data packets. The Three-Way handshake process for connection establishment is shown step-by-step below:

Step 1. The sender first sends a packet with a **SYN flag** to the receiver to establish a connection.

Step 2. The receiver responds with a packet having a **SYN-ACK flag,** acknowledging the request and indicating that it is ready to receive the data.

Step 3. The sender then sends the **ACK packet**, confirming that the connection has been established.

These 3 steps complete the Three-Way Handshake process and establish a TCP connection between two computers.

How is TCP Connection Terminated?

After the data has been shared between computers, and there is no further need for data transfer, TCP terminates the connection between computers. To terminate the connection between computers, TCP uses the **four-way handshake** process.

The steps in a four-way handshake are:

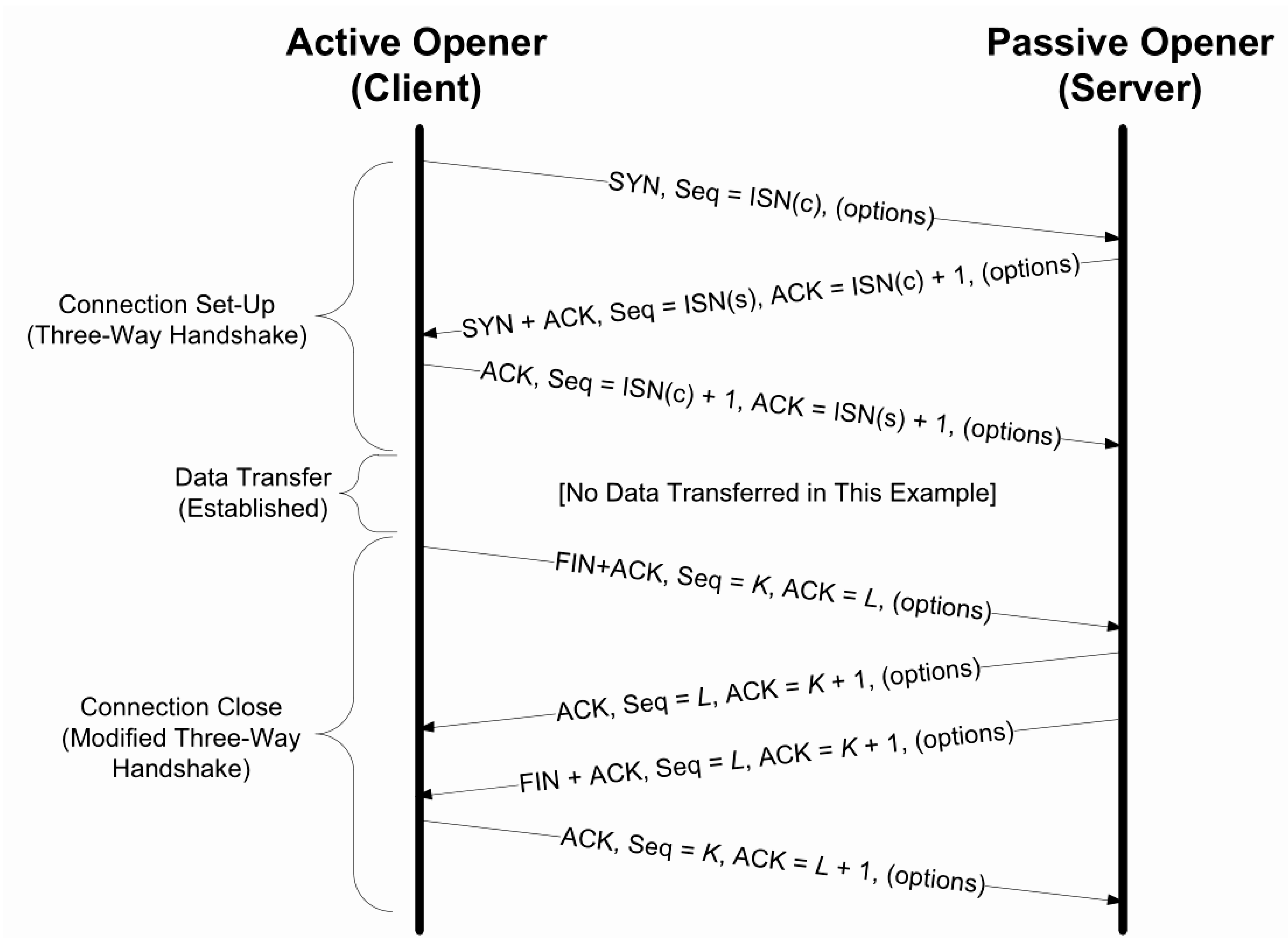
Step 1. One device sends a **FIN packet** to signal the end of the connection.

Step 2. The other device responds with an **ACK packet** to confirm receipt of the FIN packet.

Step 3. The same device then sends its own FIN packet.

Step 4. The original sender responds with an ACK packet to confirm receipt.

This completes the four-way handshake process, and the connection is closed.


Active Opener
(Client)
Passive Opener
(Server)
SYN, Seq = ISN(c), (options)
SYN + ACK, Seq = ISN(s), ACK = ISN(c) + 1, (options)
ACK, Seq = ISN(c) + 1, ACK = ISN(s) + 1, (options)
Connection Set-Up
(Three-Way Handshake)
Data Transfer
(Established)
[No Data Transferred in This Example]
FIN+ACK, Seq = K, ACK = L, (options)
ACK, Seq = L, ACK = K + 1, (options)
FIN + ACK, Seq = L, ACK = K + 1, (options)
ACK, Seq = K, ACK = L + 1, (options)
Connection Close
(Modified Three-Way
Handshake)